Sampling Process

Hello, and welcome to another session on business statistics for entrepreneurs. In the last module, we discussed continuous probability distributions. In this module, we are going to discuss the idea of sampling and sampling distributions. Joining me in this journey are our bright students Tejas and Ashwini. As always, they are going to keep me honest and on track. How are you both doing? Shall we get started? Yes, I'm ready to move on from probability distributions to sampling techniques. Yes, I am curious to know better methods to collect data. Yes, of course. Let's begin by asking, what is sampling? Sampling is the process of selecting a subset of observations from a population.

But why are we collecting this data? We collect this data to not just analyze the sample, but we see it as an efficient way to make inference about various population parameters such as the mean, proportion, and standard deviation. In this context, sample becomes an important step in inferential statistics.

Therefore, an incorrect sample may lead to wrong inference about the population. In this module, we will first discuss the difference between sample statistics and the corresponding population parameters. Then, we will discuss the importance of using appropriate sampling techniques to make reliable and trustworthy inferences about the population.

We will see various sampling techniques used in practice and the limitations of various sampling techniques. Then we will learn about sampling distributions and how we can estimate population parameters using the information about sampling distributions. Here, an important theorem called the central limit theorem will provide us with some powerful results.

Finally, we will look at some properties of point estimates that are obtained using sample data. Let's get started. To understand the difference between sample statistics and the corresponding population parameters, let's revisit some concepts that we did in the first module. If you recall, in the first module, we talked about elements, population, and sample.

We said that an element is an entity for which data is collected, while a population is the collection of all elements of interest. And the sample is a subset of this population. The reason

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we select a sample in the first place is to collect data that enables us to make inferences and answer research questions about the population. Let us look at two scenarios to understand this better.

Viren Agarwal is an analyst at the Jaipur office of a leading news channel. State elections are soon to happen in Rajasthan. All political parties are interested to know the expected outcome of the upcoming elections, especially in some hotly contested seats.

Viren has been asked to collect data to provide some estimates. He has collected data from a sample of 400 registered voters in the state of Rajasthan, and 160 out of these 400 voters indicate a preference for a particular candidate. Based on this sample data, he estimates that the proportion of the population of registered voters that are favoring of this particular candidate is going to be 160 divided by 400, which is equal to 40%.

Is he correct? Is his estimate good? Is it reliable? Not sure. 400 is a small number of voters. Not sure if this proportion reflects what will be happening in the population. It depends if this sample of 400 is representative of the population. Yes, you both made valid points. In this example, the 40 percent Viren obtained is the sample statistic, and he is using it as an estimate of the true proportion in the population, which we would consider to be a population parameter.

Let's take another example. Manoj Choudhury, who works in the HR department at a power plant in Kota, Rajasthan. Ashwini and Tejas, do you know what KOTA is famous for? Coaching classes. Entrance exam preparations. Well, you both were very quick to point that out. Anyway, back to Manoj. He has been asked to estimate the average yearly earnings of migrant temporary workers that work at the plant.

He interviewed and collected the data from 120 temporary workers that worked at the plant last year. His study revealed that their average salary is around 36, 500 per year. Based on this result, he claims that the average salary for all migrant workers in quota is 36, 500. Is he justified? Are his estimates reliable?

I'm not sure. There are so many migrant temporary workers and his sample is small. Since workers are migrant, I don't think that 120 would be a good representation of the population.

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Yes, and in this case, It is even hard to identify the composition of the entire population. Manoj's value of 36, 500 is the sample statistic and he is using it as an estimate of the true average in the population, which we would consider to be a population parameter.

In both these examples, the proportion of votes estimated by Viren for the election prediction or by Manoj for the average salary of migrant workers are only estimates of the values corresponding to the corresponding populations. The actual values in the population will most likely be different.

Because the sample only contains a portion of the population, we do know that there will be some error. But the question is, how much will the error be? Are these estimates reliable enough to make decisions and conclusions? With proper sampling methods, sample results such as these should provide fairly good estimates of what the population parameters should be.

The statistical techniques that we will see in this module will allow us to provide some guarantees and confidence in our estimates. The sampling process itself has many steps. And each step is important to ensure that a good sample is used for making inferences about a population. The first step is the identification of the target population.

This is important for any study. For example, in Manoj's case, where he was interested in identifying the earnings of the migrant temporary workers. The definition of migrant workers is vague. We would need a better definition to identify the target population better. Such a definition would put boundaries in terms of some of the parameters.

What do you think would be some of the relevant parameters? Age? Gender? How about type of work done by the workers? Yes. All of these would better define this target population. The second step is to decide the sampling frame. The sampling frame defines the source of the method or the procedure used to identify the elements of the target population.

The choice of the sampling frame is important because it could determine the accuracy of the study. For example, in Viren's case. He would be able to use the census data or the voter registration data to identify all the elements of the target population, that is, all possible voters in Rajasthan.

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However, deciding the sampling frame is not easy in Manoj's case. There is no pre built database of temporary migrant workers in Kota. This makes identifying the sampling frame much harder in his case. The third step is to determine the sample size. The sample size is a minimum number of records that we need for a study.

Surely the sample size should be as big as possible, but that is easier said than done because data collection can be expensive, both in terms of cost and time. And at the same time, an insufficient sample would result in a lack of precision in the estimation of parameters. The size of the sample could also affect the desired level of confidence and the margin of error in our estimates.

The fourth step is to choose the sampling method itself. The sampling method is the technique that is used to select the individual cases in the sample from the target population using the sample frame. At a high level, sampling methods can be classified into two categories, probabilistic sampling and non probabilistic sampling methods.

Let's see each of these in more detail.

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Sampling Methods

Let us start with probabilistic sampling. As the name suggests, in probabilistic sampling, individual observations in the sample are selected from the population according to a probability distribution. For example, in Viren's case, it is likely that the population has more than 4 crores of registered voters. And Viren is interested in sampling 400 voters. Now, there are several ways to choose 400 voters from 4 crores of voters. Even within probabilistic sampling techniques, this can take many forms. One of the forms that is commonly used is known as random sampling. Random sampling is ideal when the population is homogeneous.

In random sampling, every subject in the population has an equal probability of selection into the sample selection of cases. But how do you ensure that? The execution of random sampling can be done through several procedures. One commonly used method is to label all the cases in the population sequentially.

So in Viren's case, we would label the population from 1 all the way up to 4 crore. Then you generate uniform random numbers and select the records from the population. How do you generate uniform random numbers? Yes, we only learn how to calculate probabilities for the random variables, not how to generate them.

Yes, that's a good point. But there are several ways to do that. One simple way to do that in Excel is to use the functions RAND( ) or RANDBETWEEN( ). Both of these functions can be used to generate random numbers. Using these functions, Viren could identify a random sample of 400 voters from the list of registered voters.

Another probabilistic technique used is known as stratified sampling. In stratified sampling, the population can be divided into mutually exclusive groups. We do this using factors like age, gender, marital status, income, demography, and so on. The groups thus formed are called strata, and hence the name stratified sampling.

But there will be several ways to create these groups. Yes, that is true. And it is important that the groups that are defined are both mutually exclusive and collectively exhaustive, so that they represent the entire population. Yes, we talked about this when we did basic probability. Yes,

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and in simple terms, we should cover everybody through the groups, but not have overlap between the groups.

For example, in Manoj's study about temporary migrant workers. Strata could correspond to male and female migrant workers. We may also look at strata that is not defined by gender, but by age group of migrant workers. Workers that are less than 20 years old, workers that are between 20 and 30 years of age, workers between 30 to 40 years of age, and so on.

Once you have identified the strata, then random sampling methods that we discussed can be used within each stratum to select the individual records to generate samples for each group. Each strata is roughly homogeneous and so we can use random sampling within a strata, even if not across the entire population.

Yes, Ashwini, that's the logic. The size of the sample in each stratum should be proportional to the size of the stratum in the population. For example, if the proportion of women in the population is 40%, then we would want the proportion of women in the sample also to be 40 percent of the total sample.

Samples from each stratum are then combined to create the final sample. Seems logical, but I did not realize so much thought goes into deciding the sample. Yes, careful thought is required. Stratified sampling could be necessary if the population is heterogeneous, because if it is so, simple random sampling might not yield a sample that appropriately reflects the population.

So for Viren's case, the population of migrants is definitely not homogeneous. So it will be better off trying to look at various strata of the workers and then ensuring that the random sample reflects every strata of the group. If the elements within the strata are all alike, then the strata will give fairly reliable estimates.

And in this case, even relatively small sample size can give very good estimates of the strata characteristics. If the strata are homogeneous, then stratified sampling procedure provides results that are as precise as simple random sampling with a small sample size. The third type of probabilistic sampling method is cluster sampling.

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In cluster sampling, the elements of the population are first divided into separate groups called clusters. Each element in the population belongs to one and only one cluster. A simple random sample of clusters is taken and then all elements within each sample cluster form the sample. When do we choose cluster sampling?

Cluster sampling is helpful if the elements within the cluster are diverse and not alike. That's the difference between strata and cluster. We want strata to be homogeneous, but cluster to be heterogeneous. Yes, Ashwini, you describe the difference between strata and cluster precisely. In the ideal case, each cluster must be a small scale representation of the entire population.

The value of cluster sampling methods depends on how representative each cluster is of the entire population. If different clusters are alike and each cluster is a pretty good replica of the population, then a small number of clusters will be enough to provide results. One common way of cluster sampling is area sampling.

In variant study of water preferences. This might very well apply. Here, the clusters might correspond to districts, municipalities, or wards. If Viren identified certain municipalities or wards in Rajasthan and surveyed all of the people in those municipalities or wards, he might get a very good representation of the entire population of Rajasthan, especially if he chooses municipalities or wards that are good representation of the population of Rajasthan.

Now I see. We had representatives from a bank. Come and talk to students in our college about credit card preferences. Yes, and I guess they were using your college as a cluster to get an idea of what the entire population of college students might prefer. Cluster sampling generally requires a larger sample size than simple random sampling or stratified sampling. However, it can result in cost savings because to collect the data you just need to focus on a limited set of clusters and therefore, it might be more efficient both in terms of cost and time.

Next, we look at non-probability methods of sampling. In non-probability sampling methods, the selection of sample units from the population does not follow any probability distribution. Sample units are selected based on a systematic non-probability based method or based on convenience sampling methods. Let's first discuss systematic sampling. We use systematic sampling when it is time consuming to do simple random sampling. Why is that time

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consuming? It could be time consuming because you need to first generate random numbers and then search through the entire list of the population to find the corresponding elements.

For example, in Viren's study of the population in Rajasthan, he will have to generate 400 random numbers first and then corresponding to each of those random numbers, find the element in the database that has four core entries. Then survey each one of those elements. In contrast, in systematic sampling, if Viren wanted a sample size of 400 from his database of four core entries, He will need to sample one element for every 100, 000 elements in the population.

What Viren would then need to do is to first randomly select one out of the first 100, 000 elements from his population list. Then the other sample elements are identified by starting with this first sampled element and then selecting every 100, 000th element that follows in this population list. In effect, the sample of 400 is identified by systematically moving through the population and identifying every hundredth element after the first randomly selected element.

Now I see, this way of putting together a sample of 400 should be easier than simple random sampling. Yes, and because the first selected element is a random choice. A systematic sampling is usually assumed to have some properties of the simple random sampling method.

Next, we talk about convenience sampling. Convenience sampling is also a non-probability based sampling technique. As the name implies, the sample is identified primarily by convenience. Elements are included in the sample without pre specified or known probabilities of being selected. For example, if Manoj wanted to survey the temporary migrant workers, he may just use the migrant workers that he has interacted with at his plant in the last month because they are readily accessible and might agree to participate as subjects for little or no cost.

I can think of an example. Traffic inspectors pulling out motorists at an intersection to verify if their license and registration documents are in order. Yes, Tejas, that's a good example. As you can imagine, it would be complicated to identify motorists that need to be surveyed using a probability distribution.

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It is much easier to pull out the next motorist driving by the intersection for the check, and once the inspector is done inspecting the documents, of a motorist, he or she pulls the next motorist driving by. Convenient samples have the advantage of being fairly easy in terms of sample selection and data collection.

However, it is impossible to evaluate the goodness of the sample in terms of how well it represents the population. A convenient sample may provide good results or it may not. There are no statistically robust procedures available to be able to conduct a probability analysis and make robust inferences. Therefore, we should be cautious about interpreting the results of convenient samples that are used to make inferences about the population.

Judgment sampling is another approach where a person that is most knowledgeable on the subject, selects elements of the population that he or she feels are most representative of the population. Is this like the panel we see on televisions that express their opinion on a political issue? Yes, it is. And often this method is a relatively easy way of selecting a sample. For example, Viren could easily ask a handful of people from a municipality or ward and argue that these people reflect the general opinion of the entire population.

However, the quality of the sample results depend on the judgment of the person selecting the sample. How reliable are Viren's choice of people used for the study? Therefore, we need to be very careful in drawing conclusions based on judgment sampling if it is used to make inferences about the population. Another non-probability based sampling method is known as voluntary sampling. In voluntary sampling, data is collected from people who volunteer for such data collection. A good example is the customer survey requests by companies and service providers. Of course, there would be bias in the case of voluntary sampling.

We know that primarily customers with bad experiences are the ones that respond to such survey requests. In most cases, customers with good experiences do not bother to fill in such survey requests. Another example is clinical trials conducted by pharmaceutical companies. During this process, the company asks for volunteers for a particular study.

Although there is some effort to choose a representative subset from the set of volunteers, there is no guarantee that the final choice is a good representation of the population.

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Analysis of Sample Data

Now that we have described various methods of sampling to collect data, let's now talk about the next task, which is the analysis of the sample data. One of the main purposes of this analysis is to is to get estimates for the parameters related to the population. The most common population parameters of interest are the population mean, usually denoted by mu, the population standard deviation, usually denoted by sigma.

Very often, we are also interested in something known as the population proportion, P, the proportion of people in the population that satisfy some criteria of interest. To help us understand this process of estimation, let us revisit two examples that we saw earlier in module one.

In the first module, we had looked at the data that Basavaraju had collected from his customers of the chroma store in Ballari. He had collected data related to customer satisfaction. Tejas, do you remember that? Did it have something to do with computer purchases and satisfaction? Yes, you are right. Here is the data again. Basavaraju had 36 data points corresponding to 36 survey respondents. In his data, he had recorded the brand of laptops that they had purchased, the price that they had paid for the laptops, some features of the laptops that they had purchased, such as RAM, hard disk, and operating system.

And finally, he had asked them for a satisfaction score between 0 to 100. This score reflects the satisfaction level related to the store or purchasing experience. When we looked at this data earlier, we had looked at the types of variables, quantitative and qualitative. We had also looked at which of them were measured on a nominal scale, ordinal scale and so on.

We had also computed their descriptive statistics. Here is what we had computed. For our discussion in this module, we will focus mainly on the descriptive statistics related to the satisfaction score. We have 36 points in this data set. We calculated the average of the 36 points to be equal to 57.31

ashwini, what were the median and the mode? The median turns out to be 59 and the mode is 61. And Tejas, what do the mean, median, and mode tell you? They are both higher than the

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mean, which means there may be some high satisfaction scores. And any concern that there might be some outliers? Perhaps not. We can check.

But the mean and median are close enough. And Ashwini, what about the spread? The range is between 40 to 266. So not too much spread. And the standard deviation is 6. 38. Roughly 11 percent of the mean. Yes. And Basaraju is also interested in the proportion of customers that gave a score of 60 or more. He considers a score of 60 out of 100 to be a score indicating reasonably satisfied.

If we count how many of his 36 customers gave a score of 60 or more, it comes out to be 17. So the proportion would be 17 divided by 36, which is around 47%. As we said earlier, the main purpose behind Basaraju collecting the data is not to understand what 36 of his customers feel, but to really use what they have reported to infer what it says about the satisfaction level of all his customers, even those whom he did not collect the data. He wants to know what this average score of 57. 31 indicates about the population mean mu, the average satisfaction score corresponding to the population. Does the same idea hold for standard deviation?

Yes, indeed. He wants to know what the standard deviation of the score of 6. 38 indicates about the population standard deviation, sigma, the spread in the satisfaction score corresponding to the population. Similarly, for proportion 2. He wants to know what this proportion of 47 percent of customers giving a score of 60 or higher and acknowledging to be reasonably satisfied means in terms of the proportion P in the population of people that are reasonably satisfied.

Sometimes you will hear data scientists and others use the word sample statistics. What are they? Are these the values we computed from the sample? Yes, given this larger goal related to population parameters, the corresponding characteristics that Basaraju has obtained as descriptive statistics are often referred to as sample statistics.

For example, to estimate the population's average score mu, Basaraju has obtained the corresponding sample average x bar to be equal to 57. 36 using the formula we learned earlier. Similarly, to estimate the population standard deviation sigma, Basaraju has obtained the corresponding sample standard deviation S to be equal to 6. 38. And to estimate the population

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proportion P, the proportion of customers that are reasonably satisfied in the population, Basaraju has obtained the corresponding sample proportion P bar to be equal to 0. 47.

In statistical inference, there is also the term point estimates for parameters. One would say that in making the computations that we did here, we followed the statistical process of getting point estimates. We refer to the sample mean x bar of 57. 34 that we just calculated to be the point estimator of the population mean mu. Similarly, The sample standard deviation S that we just computed to be equal to 6. 38 is considered to be a point estimate of the population standard deviation sigma. And the sample proportion P bar of 0. 47 that we just computed of reasonably satisfied customers is a point estimator of the population proportion P. The numerical values are each called point estimates of the corresponding unknown population parameters.

But these estimates are not unique. No, they are. How many other averages can you get from the 36 values of scores? Well, you are both right in a sense, whether they are unique or they are not. It is important to realize that our x bar of 57. 31 is but one out of many possible point estimates for the unknown population mean, mu.

After all, we have only one survey result that Basaraju has collected, and given that survey, this would be the only estimate of x bar. But, if Basaraju, or you, or I, did a similar survey on the same population, we will perhaps get a different estimate x bar for the same unknown average mu in the population.

Similarly, our standard deviation S of 6. 38 is only one out of many possible estimates for the unknown population standard deviation, sigma. Again, because we are deriving it from the one single survey that Basaraju has collected, and so therefore, this is the only estimate of standard deviation we could get.

However, if Basaraju or you or I did a similar survey again on the same population, we would perhaps get a different estimate as for the same unknown standard deviation sigma in the population. The same holds for the population proportion 2. Our sample proportion p bar of 0. 47 is only one out of many possible estimates for the unknown population proportion p. That is because we are deriving it from the one single survey result that Basaraju has given us. If we

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had another survey on the same population, we would get perhaps a different estimate p bar for the same unknown population proportion p in the population.

Of course, point estimates would differ somewhere from the actual population parameters. This difference is to be expected, because after all, it is a sample and not derived from the entire population. But the main intent here is to see how we can construct an interval of confidence using the available information. That tells us how close the point estimates that we have collected is to the population parameters that are of interest to us.

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Sampling Distribution

Let's understand this notion of point estimator in a little bit more detail. We said that the sample mean x bar is a point estimator of the population mean mu. We said that the sample proportion p bar is a point estimator of the population proportion p. And the sample estimate S is a point estimator for the population parameter sigma.

For the survey that Basavaraju conducted of his 36 customers, we got the point estimate of mu to be equal to x bar equal to 57. 31. We got p bar equal to 0. 46, the point estimate of P, of customers who gave a score of 60 or more. And ‘’we got the point estimate of sigma to be S equal to 6. 38. Now, suppose we had selected another sample of 36 customers then a different value of x bar, p bar, and s would have been obtained. For example, Tejas, if you had conducted a survey of Basaraju's customers, you might get the value of x bar to be equal to 56. 3, the value of p bar to be equal to 0. 45, and the value of s to be equal to say 6. 71.

And then let's say Ashwini, you did a survey of Basavaraju's customers, you might get a value of x bar to be equal to, let's say 58. 02, the value of the proportion p bar to be 0. 48, and the value of the standard deviation s to be equal to 6. 13. So now suppose we repeat this process of selecting a sample of 36 customers over and over again, and each time we compute the values of x bar, p bar, and s. We might get values that fill a table like this one shown here, with the results obtained from 500 such simple random samples.

Notice that in each case we are sampling from the same population, and each one of these estimates of x bar is a different point estimate of the same population mean, mu, and each one of these estimates, p bar, is a different point estimate of the same population proportion, p, and each one of these estimates, s, is a different point estimate of the population standard deviation, sigma.

It is like tossing a die and observing the value that we get. Yes, Ashwini, it is actually interesting that you should say that. But I'm going to pick up on what you ;just said in a minute here. Let me finish what I was getting at. So if, and that's a big if, if we had the luxury to repeat the data collection several times and get all these different values of x bar, p bar, and s, then,

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we can plot the frequency and the relative frequency distribution of these 500 values of x bar, the 500 values of p bar, and the 500 values of s.

Yes, we can plot them. We will get a histogram. But what do we call it? I mean, what would we be looking at? That's a good question, Tejas. If you recall, we had defined the random variable as a numerical description of an outcome of an experiment. Yes, the probability is the likelihood of the outcome the random variable represents.

Exactly. Now, if we consider this process that we just did, the process of selecting a simple random sample from a population as an experiment, Then, the sample mean X bar is a numerical description of the outcome of that experiment. And therefore, the sample mean X bar is a random variable. Continuing my die analogy, if you toss a die 500 times and you get values of 3, 1, 6 and so on, you conduct survey 500 times and you get values of X bar to be 57. 31, 56. 63, 58. 02, and so on. Yes, Ashwini, your analogy is appropriate here. So in that sense, Basaraju is conducting a random experiment when he is collecting sample data and recording the outcomes of interest, x bar in this case, except he has only one outcome, like tossing a die once. Let's just focus on X bar for the time being and build on this idea that X bar is a random variable.

What do we know about random variables? They are either discrete or continuous. Yes, and what else is true about all random variables? They will have a PMF or a PDF that connect values of the random variables to probabilities. Yes, they all have a probability distribution. In this case, X bar, P bar and S will all be continuous random variables as they can take fractional values. Yes, that is right. Now, what is the first thing we try to do when we have the probability distribution for a random variable? Find cumulative probabilities, tail probabilities, inverse probabilities. Yes, and? Find its mean or expectation and its variance. Yes, that is true.

That is what we've been doing for all the random variables that you've discussed so far, right? So we could do the same for x bar also, correct? And that implies that our random variable x bar should have a PDF and the plot of these 500 values of x bar would be an approximation of this pdf. And I say approximation, because it is based on just a finite number of 500 values.

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The actual probability distribution of x bar is known as the sampling distribution of x bar. So Tejas, to answer your earlier question about what plot we are looking at, the plot of that histogram of 500 values of x bar would be an approximation of the sampling distribution. What will its shape be?

Something we have seen or something new? And how does knowledge of this sampling distribution of x bar help? Seems like you are getting curious and anxious. Suspense is building up, right? Patience. We will answer all of those questions. Knowing the sampling distribution of X bar, its name, and its properties, like the expectation and the variance, will allow us to make some useful probability statements about how close the sample mean X bar is to the unknown population mean mu.

Oh, wonderful. So we need Basav Raju to repeat the data collection 500 times to make a connection between X bar and mu. No, not really. Now, in practice, we select only one simple random sample from the population. Just like what Basaraju did. Basaraju neither has the luxury of time or the budget to be able to repeat this data collection multiple times, let alone 500 times.

We imagine this hypothetical case of repeated sampling by Basaraju only to illustrate that many different samples are possible and that the different samples generate a variety of values of the sample statistics x bar. However, we know that this one outcome is an instance of a value from a probability distribution that we will call as a sampling distribution.

And with the sampling distribution, We can make the required connection between x bar and mu. So the x bar value of 57. 31 that Basav Raju for is one instance for a value that could come from sampling distribution of x bar. Yes. And the value of p bar equal to 0. 47 is one instance of a value. That could come from the sampling distribution of P bar.

True again. And the value of S equals to 6. 38 is one instance of a value that could come from a sampling distribution of S. Yes, that's true. And you are going to tell us the names of all these distributions and make the connections with the relevant population parameters? Yes, it seems like you want the answers right now.

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Yes, you have made me curious. From one value of x bar, I can say something of mu. Yes, indeed. So let's see the details of the sampling distribution of X bar. We said that the sample mean X bar is a random variable and its probability distribution is called the sampling distribution of X bar. This probability distribution, the sampling distribution of X bar, is the probability distribution of all possible values of the sample mean X bar.

We will now look at some of the properties of the sampling distribution. Just as with any probability distribution that we have studied so far, the sampling distribution of x bar also has an expected value. It also has a standard deviation. It also has a shape or a form. Let's begin by looking at the expected value, that is the mean of all possible values of x bar.

We will denote this expected value of x bar by E of x bar. Recall that we have used the symbol mu to represent the mean of the population from which we are selecting these simple random samples. Then, using the results that we had discussed earlier for linear combinations of random variables, It can be shown that for a simple, random sampling, the expected value of X-bar or E of X-bar is equal to mu.

In other words, the expected value of X-bar equals the mean of the population from which the sample is selected when the expected value of a point estimator X bar in this case. equals the population parameter, mu, in this case, then we say that the point estimator is unbiased. Thus, since E of x bar is equal to mu, x bar is an unbiased estimator of the population mean, mu.

How does that help in determining mu from x bar? x bar need not to be mu, right? 57. 42 is not necessarily mu. That's true, Ashwini. x bar is not necessarily mu. But although Basavaraju might have got only one value of 57. 3, one which may not be necessarily the mean of the population, however, we can be rest assured that this value that he got comes from a distribution that is centered in a good place.

It is centered at exactly the population mean mu, that we are trying to determine. And you will see that this will be useful to determine mu. How far our single value of x bar is from the mean or the expectation, of course, will depend on the spread of the distribution of x bar. So now we talk about the standard deviation of the sampling distribution of x bar.

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Of course, if the standard deviation of the sampling distribution of x bar were zero, then our x bar value would always be the expectation, and we are done. But that is too good to be true. So let us call sigma x bar as the standard deviation of x bar, to distinguish it from sigma, the standard deviation of the population.

Now, in textbooks you will find that the expressions for sigma x bar are there for two cases. One is where the size of the population, capital N, is finite, and the second is the case where the size of the population, capital N, is considered to be large enough or close to infinity. When the sample size, which we are denoting by small n, is comparable in order of magnitude, for instance, to capital N, then the expression for sigma x bar requires some adjustment.

However, I am not going to get into those details. Instead, I am going to assume that in all the cases that you are concerned about, the population size capital N is large enough compared to our sample size small n that we can use the expression for sigma x bar that assumes that the population is actually infinite.

In this case, it can be shown that the standard deviation of x bar is sigma x bar and is given by the sigma divided by root n. Note that to compute sigma x bar, we need to know sigma, the standard deviation of the population. From a notation and terminology perspective, to emphasize the difference between sigma x bar and sigma, we refer to the standard deviation of x bar, sigma x bar, as standard error.

The term standard error is used throughout statistical inference to refer to the standard deviation of a point estimator. The main reason for this new term is so that we do not confuse it with the standard deviation of the population, but it plays a role that is very similar to standard deviation. In this case, for our point estimator x bar, we will soon see that the value of the standard error x bar will help us determine how far Basavaraju's sample mean x bar of 57. 3 is from the population mean mu. Notice a few things though. If the sample size increases, the standard error decreases. So, if we had collected a large sample and was not 36, but much higher, then our x bar would be fairly close to the population mu. That is why we like large sample sizes. Yes, that is one of the reasons.

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Large sample sizes have several benefits, but it also comes with a burden of cost and effort. Let me go back to the die example to illustrate. Tejas, what is the average value you get when you toss a die, an unbiased die? You mean for the discrete uniform distribution? Yes. It is 3. 5, right? Yes, and imagine that is our population mu, except in this case we know that mu is equal to 3. 5. Now, let's say you toss a die twice. and you get values of 1 and 2. So, your x bar is 1. 5. And if Ashwini tosses the die twice, let's say she gets values of 5 and 6. So, her x bar is 5. 5. Now, both these values are far from the true mu of 3. 5, because your sample size is small, n is equal to 2. But now, if you both toss this die a hundred times, you both might get different values of x bar.

But they will both be fairly close to the true value of 3. 5. Yes, I see that. But can we say how close? We will answer that pretty soon. The second point about the expression of x bar equal to sigma divided by square root of n is that it requires sigma, the standard deviation of the population. Oh yes, I never realized that.

We may not even know mu, let alone sigma of the population. Yes, and that is true in many cases. But we will address that issue in more detail as we go forward. But for now, let's say that Basaraju knows sigma for the satisfaction score. Let's say he has estimated sigma when he did the survey last year.

And let's say it was 6. And he's okay with assuming that value of sigma for now. Then the standard deviation sigma x bar is going to be equal to 6 divided by the square root of n or 6, divided by the square root of 36, which would be equal to 1. So, where are we? We know that Basavaraju's x bar of 57. 3 comes from a probability distribution centered at the population mean mu, and it has a standard error equal to sigma divided by root n, which is 1, if we assume the sigma of the population to be 6, and our n of 36.

The final step in identifying the characteristics of the sampling distribution is to determine the form or the shape of the sampling distribution. In order to do that, we need to remind ourselves of what we know about linear combination of random variables. In particular, the results related to the expectation and the variance of the average of n iid random variables.

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Let's say we have n independent and identically distributed, that is IID, random variables, each coming from the same distribution with mean mu and standard deviation sigma. Then, Tejas, what is the expectation of their average x bar? We have to apply that a1 times expectation of x1 plus a2 times expectation of x2 formula, right?

Yes, and if you did that, what would you get? Expectation of x bar is mu. Yes, and what is the variance and the standard deviation of x bar? The variance would be x bar divided by root n and standard deviation is sigma divided by root n. Oh, that is the result we used to get the values for the expectation and standard deviation of x bar a few moments ago. Yes, that is right. After all, in Basaraju's case with n equal to 36 and each of his 36 values of the scores, are iid random variables or samples drawn from the same population with some distribution with mean mu and standard deviation sigma. Now, this holds regardless of the distributions of the xi's, right?

Yes, it is just linear combination. So it will hold regardless of what the population distribution is. Okay. Now, if we know that the xi's are normal with mean mu and expectation sigma, then what do we know about the linear combination of normals? They would be normal too. Yes, and so x bar would now be coming from a normal with mean mu and standard deviation of sigma divided by root n.

In other words, if the population has a normal distribution, then the sampling distribution of x bar is normal with the expectation equal to mu and the standard error equal to sigma divided by root n.

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Central Limit Theorem

What happens when the population from which we are selecting the random sample does not have a normal distribution. First, realize that the expectation of x bar will be mu of the population's distribution, and the standard error of x bar will be sigma divided by root n. The real question is what is the distribution?

We know it is some distribution with the expectation and the standard error that we just mentioned, but what distribution? Here is where the central limit theorem gives us an important result. The central limit theorem is helpful in determining the shape of the sampling distribution of x bar. The statement of the central limit theorem as it applies to the sampling distribution of x bar says the following.

When you select random samples of size n from the population, the sampling distribution of the sample mean x bar can be approximated by the normal distribution if the sample size becomes large. It is normal regardless of the population distribution. No, it can be approximated by the normal if the sample size is large enough.

Okay, it can be approximated by a normal. Yes, let's see a picture of what is happening. This figure shows how the central limit theorem works for four different populations. Each column refers to one of the populations. The top panel shows the population distribution of x in the population. In Basaraju's case, remember the distribution of the x is the distribution of the scores in the population which has mean mu and standard deviation sigma. In the panel, the first picture corresponds to the normal. The second is uniform. The third is skewed like the exponential. And the fourth one is some arbitrary distribution. Let's call it the purple distribution. The bottom panels show the shape of the sampling distribution of X bar.

The sample averages that we know has mean mu and standard error sigma divided by root n. When the sample size n is equal to 2, the shape of the distribution X bar is not the shape of the population, nor is it the shape of the normal. But when n increases to beyond 30, we see that the shape of the sampling distribution of x bar begins to look very similar to the shape of a normal distribution.

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Is it 30 always? Why at n equals to 30? What is so special? From a practical standpoint, we want to know how big the sample size should be before the center limit theorem approximation is good enough. Several researchers have investigated this question by studying the sampling distribution of x bar for a variety of populations and a variety of sample sizes.

The general practice is to assume that for most applications, the sampling distribution of x bar can be approximated by a normal distribution whenever the sample size is 30 or more. Note that n is equal to 30 or more is an empirically derived number that works in most cases. While it is theoretically proved that the distribution of x bar converges to the normal as n goes to infinity, it has been empirically observed that even at n equal to 30, the distribution of x bar starts to look like the normal in most cases.

Of course, if you want, you can always find distributions for the population, especially ones with high skewness. Where even with sample sizes of n more than 150 or more, the sampling distribution may not appear to look like the normal. But this does not happen for most distributions of the population.

But how come the population distribution becomes normal for large samples? No Tejas, I never said the population distribution becomes normal for large samples. Isn't that what Central Limit Theorem says? No, let me repeat what it says. When you select random samples of size n from the population, the sampling distribution of the sample mean x bar can be approximated by the normal distribution if the sample size becomes large.

The central limit theorem is talking about the distribution of x bar and not of the population. Yes, and that is an important point. We are talking about the distribution of X bar and not the distribution of X in the population. Clearly, just because you sample more than 30 from a population, the distribution of the population cannot change.

However, the shape of the distribution of X bar, the random variable, starts to look more like the normal. Okay, I think I get it. It will take some time to absorb this. Yes, this requires some thinking through about what is it that the Central Limit Theorem is saying and what it is not saying. But let's see what the Central Limit Theorem or CLT as it is often called in short, let's see what it means for Basavaraju's survey.

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First, Basavaraju realizes that he cannot expect the value of the sample mean x bar that he has found to be equal to 57. 31 to be exactly equal to the population mean mu. The reason we are interested in the sampling distribution of x bar is so that it can be used to provide some probability information about the difference between the sample mean x bar and the population mean mu.

We have learned so far that the sampling distribution of x bar is normal and Basaraju has a sample average x bar equal to 57. 31 and the sample standard deviation to be equal to 6. 38 for a sample size of 36. Now, let's say that Basaraju believes that the sampling process and the sample mean estimate will be really trustworthy for his own decision making.

If the sample average x bar is within plus or minus two of the unknown population mean mu, however, it is not possible to guarantee that the sample mean x bar will always be within plus or minus two of the population mean mu because x bar is a random variable. So instead, we will determine what is the probability that the sample mean x bar computed from the survey of n equal to 36 customers will be within plus or minus 2 of the population mean. Because n is equal to 36 and it is larger than 30, we can apply the central limit theorem to say that the sampling distribution of x bar is approximately normal. Further, we know that the expectation of x bar is mu, and we also know that the sigma x bar, the standard error, is sigma divided by root n, where n is 36.

Now, if we go with Basaraju's estimate of sigma from last year's survey, what is the value of sigma x bar, Tejas? Sigma X bar is 6 divided by 6, or 1. Yes, Then we want to determine what is the probability that the normal random variable x bar minus mu lies within plus or minus 2. To calculate this, we transform this question into a probability question related to the standard normal random variable by dividing everywhere by the standard error of x bar, sigma divided by root n, which coincidentally is 1 in this case. So, what do we want? It seems we just want the probability of a standard normal taking values between minus 2 and 2. Yes, and that probability is 0. 95. Thus, there is a 95 percent chance that Basaraju's x bar of 57. 3 is within plus or minus 2 of the population mean.

Alternatively, there is only a 5 percent chance that this difference between x bar and mu will be more than 2 units. That is interesting. With just one sample of 36 responses, he is able to

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come up with a pretty good estimate of population mu that it is within plus or minus 2 of his 57. 31, 91 percent of the time.

Imagine what would have been the case if he had got 100 responses instead of 36. Why imagine? Let's compute what would have been the case if n was indeed 100. Suppose that Basavaraju had obtained responses from 100 customers instead of the 36 responses that he got. Intuitively, it would seem that with more data provided by the larger sample size, The sample mean based on an n equal to 100 should provide a better estimate of the population mean mu than the sample mean obtained based on n equal to 36.

Question is, how much better? What is the relationship between the sample size n and the sampling distribution of x bar? Note that the expectation of x bar, E of x bar, is equal to mu regardless of the sample size. It does not change with sample size n. This means the x bar would always come from a distribution centered at mu, no matter how large or small the value of n is.

Yes, however, the standard error of the mean sigma x bar is given by sigma divided by root n, and it has the square root of the sample size in the denominator. So, whenever the sample size n is increased, the standard error of the mean sigma x bar will decrease. With n equal to 36, the standard error of the mean was 6 over the square root of 36 equal to 1, if we used Basavaraju's last year's estimate of sigma of 6.

However, with the increase in sample size to n equal to 100, the standard error of the mean will decrease to 6, divided by the square root of 100, which would be 0. 1. Both the sampling distributions of x bar with n equal to 36 and n equal to 100 are shown here. They are both normal centered at mu. Yes, but because the sampling distribution with n equal to 100 has a smaller standard error, the values of x bar will have less variation and will tend to be closer to the population mean than the values of x bar when n is equal to 36.

So now let's repeat our calculation to find the probability that our sample mean X bar lies between plus or minus 2 of the population mean when n is actually equal to 100. So Tejas, what is the probability that we need? We want probability of a normal random variable X bar minus mu taking values between plus or minus 2.

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Yes, and if we translate it into the corresponding standard normal. What do we get? We want the probability of a standard normal random variable taking values between minus 2 divided by 0. 6 and plus 2 divided by 0. 6. Yes, and that comes out to be the probability that Z lies between minus 3. 33 and plus 3. 33. This probability is greater than 0. 999. So, there is a 99. 9 percent guarantee that if Basaraju had conducted the survey and got 100 responses, other things remaining the same, his sample mean would have been within plus or minus 2 of the population mean. And there is only a 0. 1 percent chance that it lies outside the plus or minus 2 range.

It is really interesting that as the sample size increases, the standard error of the mean decreases. It is intuitive, but it is good to see the effect of n, so the square root of n, in the denominator of the standard error. Yes, and it is and as a result, the larger the sample size, the higher the probability that the sample mean you obtain from a single survey will be within a specified distance of the population mean.

Let's now do a few examples to get a better feel of what we just covered.

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Examples of Central Limit Theorem

Kailash Sharma works at the Secretariat in Jaipur. He is studying the unemployment data for Rajasthan. He notices that the average number of weeks an individual is unemployed is 17. 5 weeks. He wants to understand the reasons better to guide government driven remedies. From previous studies, he knows that in the population of all unemployed individuals, the mean length of unemployment is 7. 5 weeks and that the population standard deviation is 4 weeks. Kailash Sharma is proposing a follow up study where he will select a random sample of 50 unemployed individuals. Before initiating this study, he wants to know what is the probability that this sample of 50 unemployed individuals used for this detailed study will provide a sample mean that is within one week of the population mean.

He also wants to know what is the probability that this sample of 50 unemployed individuals will provide a sample mean that is within half a week of the population mean? Shall we try and answer these questions? Tejas and Ashwini, we first want to find out the sampling distribution of X bar.

Since he is selecting these 50 individuals from the population at random, X bar would be a random variable. We don't know the distribution in the population. But can we apply Central Limit Theorem, Ashwini? Yes, since his sample size is 50, we can apply CLT. Yes, and Central Limit Theorem or CLT would say that regardless of the nature of the probability distribution of unemployment duration in the population, we can say that the distribution of x bar is approximately normal.

Normal with a mean equal to the population mean of 17. 5 weeks. The standard deviation of the sampling distribution which is also known as the standard error would be given by sigma x bar, which is equal to sigma divided by root n. In this case, the sample size n is equal to 50 and the population standard deviation sigma is 4 weeks.

Therefore, what does the standard error come out to be? The standard error comes out to be sigma x bar would be 4 divided by square root of 50, which is 0. 57 weeks. Yes, and so now Kailash wants to know what is the probability that this sample of 50 unemployment individuals will provide a sample mean that is within one week of the population mean mu.

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Now, like we did for Basaraju, we want a probability that x bar minus mu is within plus or minus one. So what's next, Ashwini? We divide by sigma x bar to convert this into a probability question related to the standard normal. Yes, we divide all through by the standard error. To convert this into a probability question that is related to the standard normal random variable, what do we get? We want a standard normal random variable z to take values between plus or minus 1. 77. Correct. So we go to our standard normal template and we find that the probability to be equal to 0. 923 or a 92 percent chance.

Next Kailash Sharma wants to know what is the probability that this sample mean x bar will be within half a week of the population mean. So we want x bar mu to be within plus or minus 0. 5 weeks. Yes. Again, we translate this into a probability question corresponding to the standard normal random variable by dividing throughout by the standard error.

We get that we want z to lie between plus or minus 0. 5 divided by 0. 57, which is 0. 88. We go to our standard normal template and we find that probability to be equal to 0. 62 or a 62 percent chance. I have a basic question. Yes, Ashwini, what is it? Here, we already know the mean and the standard deviation of the population.

Then why should Kailash do a random sampling? Yes, that is a good question. Why are we doing sampling at all when we already know the mean and the standard deviation of the population? So far, when we've been talking about sampling being done. It has been done in order to estimate an unknown mean or an unknown standard deviation or an unknown population parameter.

But sometimes sampling can be done even if we know the population parameter. Take this case for instance. Kailash already knows that the mean and the standard deviation of the length of unemployment in the population. The reason for him to do a sampling study is to conduct a detailed analysis. Perhaps, the government wants to understand the challenges faced by unemployed individuals to better design government interventions and policies.

Since it will be hard to do this detailed study on the entire population, Kailash prefers to do this detailed study on a smaller population. But for the results obtained from this study on the

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smaller population to be generalizable Kailash has to make sure that the small sample is representative of the larger population.

By making sure that the sample mean is within one week of the known population mean, Kailash is setting up some assurances that the study results will really reflect the real issues faced by the larger population. I never thought of sampling being done in such a context, but this definitely makes sense.

Yes. There are several instances where sampling is done, even though the population parameters are known. Usually, the purpose is to do a more detailed study that could be reflective of the entire population. Let's look at another example. Shanti has rainfall data in Rajasthan for us. Shantibai is based out of the meteorological office in Jalawar, Rajasthan.

Shanti has rainfall data in Rajasthan for us. Shanti She is studying rainfall in various districts of Rajasthan, trying to understand if global warming has impacted the amount of rainfall received. She has found that Jalawar district in Rajasthan has the highest annual rainfall, an average of 66 centimeters per year. And Jaisalmer receives one of the lowest amount of annual rainfall, an average of 29 centimeters per year. She knows that the standard deviation of the rainfall received in both districts is 4 centimeters. To study the patterns of rainfall received by both districts more closely, she selects a sample of 30 years of rainfall data for Jaisalmer and 44 years of rainfall data for Jalawar. She wants to know what is the probability that the sample mean is within one centimeter of the population mean for the Jaisalmer district. She also wants to know what is the probability that the sample mean is within one centimeter of the population mean for the Jalawar district.

Let us help her out. Here too, Shanti knows the mean and the standard deviation of the population. And Shanti is doing this sampling study to get into more depth in terms of the rainfall patterns in both of these districts. First, we need to note that Shanti's sample would give a sample average corresponding to the average rainfall observed in the sample that she has selected.

This sample average would be a random variable. In the case of both of the districts for Jalawar and Jaisalmer, the sample sizes are 30 or more. Therefore, we can apply the central limit

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theorem. and assume that the distribution of the sample averages is going to be approximated by a normal distribution.

These normal distributions would be centered at the means that we have been given for both of the districts, and they would have a standard error sigma x bar corresponding to sigma divided by root n. Where sigma is the population standard deviations that Shanti knows to be 4 for both, and n is the number of years of data that she's using for each district.

Let's start by analyzing this for Jaisalmer. The sample size n for Jaisalmer is 30 since Shanti is collecting 30 years of data. Yes, that's right. And we know that the population mean is 29 centimeters per year for Jaisalmer. We also know that the population standard deviation is 4 centimeters per year. So now we find the standard error for sigma x bar corresponding to the Jaisalmer sample. This would be sigma divided by root n, which would be 4 divided by square root of 30 which would be 0. 73. Yes. Now, Shanti wants to know what is the probability that the sample average would be within one centimeter of the population average.

So, we want the probability of X bar minus mu to being within plus or minus 1. To convert this into an equivalent probability statement about the standard normal random variable, we divide throughout by sigma x bar or 0. 73. We need the probability that z lies between minus 1. 37 and plus 1. 37. From the template, we see that this is 0. 83. Yes.

Now next, let us repeat this calculation for the Jhalawar district. The sample size n for Jhalawar is 44 since Shanti is collecting 44 years of data. Yes, and we know that the population mean is 66 centimeters per year in Jhalawar. We also know that the population standard deviation is 4 centimeters per year.

Therefore, the standard error sigma x bar corresponding to the Jhalawar sample is 4 divided by the 44, which would be 0. 60. Now Shanti wants to know, what is the probability that the sample average would be within one centimeter of the population average? So we want the probability of X bar minus mu being within plus or minus 1 as before.

To convert this into an equivalent probability statement about the standard normal random variable, again we divide throughout by sigma X bar or 0. 60 in this case. We need the

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probability that Z lies between minus 1. 67 and plus 1. 67. From the template, we see that this is a 0. 91. Yes, and both of these samples, we yield a sample average that is close to the population mean.

And so, in that sense, they will both be very reliable. Where are we going next? We are meeting Motilal Oswal in Udaipur, who has a tax consulting practice. Motilal Oswal runs a tax consulting practice based out of Udaipur. He saw a TV report that says that the average fee charged for tax preparation for an individual by certified tax consultants in the state of Rajasthan is 1800 rupees.

The report also states that the standard deviation for the fee charged to an individual is 50 rupees. Now, Motilal is curious to know whether the fees that he has been charging his clients are representative of what is being charged in the state in general. He asks his staff to choose a sample of 40 tax returns from his past clients and compute the average fee charged. He wants to know the following. What is the probability that the mean tax preparation fee for this sample of 40 tax returns will be within 10 rupees of the average of the 1800 stated in the report? Also, if instead he looked at a sample of 75 tax returns, how will this probability change? The sample average X bar computed from a random sample of 40 tax returns of his past clients we know will correspond to a random variable.

And since the sample size is more than 30, we can apply the central limit theorem and say that the probability distribution of X bar would be a normal distribution. Yes, centered at the population mean of 1800 with a standard deviation of sigma divided by root n, where sigma is the standard deviation of the population and is equal to 50 in this case and n here is equal to 40.

So the standard error sigma x bar would be 50 divided by the square root of 40 which comes out to be equal to 7. 91. Now, we want the probability that this sample average would be within 10 of the population mean. That is, we want x by minus mu to be equal to plus or minus 10. Yes, and as we did in the previous examples, we divide by the standard error.

By doing so, we get an equivalent question related to the standard normal random variable. What do you get when you do this, Ashwini? We need the probability that the standard normal

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random variable Z lies between minus 1. 26 and plus 1. 26. We compute this probability from the template and we get it to be equal to 0. 792 or a 79 percent chance. Next, we want to know how this probability would change. if a larger sample of 75 was collected. Of course, the sampling distribution of x bar would still be normal, but now with a standard error given by sigma over square root of 75 instead of sigma over square root of 40. Therefore, the new standard error would be equal to 5. 8.

Now, we want the probability that x bar minus mu would lie within plus or minus 10, knowing that x bar comes from a normal distribution with a mean equal to 1800 and a standard error equal to 5. 8. We convert this into an equivalent question related to the standard normal random variable by dividing by the standard error throughout. And what do we get Tejas when we do this? We need the probability that z lies between minus and plus 1. 73. We compute this probability from the template and what do you get? We find it to be equal to 0.916 or 92%. Yes.

So, okay. Now we have answered Motilal's question, but now let me ask you a question. If I asked you what is the distribution of the fee charge for tax preparation for an individual by certified tax consultants in the state of Rajasthan? What is it? It is normal. Yes, it is normal with a mean of 1800 and a standard deviation of 50. Why do you say it is normal? Because the sample size is greater than 30. It can be approximated by a normal. But wait a minute. Realize we are talking about the distribution of the population here. The fee charged by all tax consultants, not the sample.

Oh, that's right. I forgot. Yes, the TV report says that in the population, the mean is equal to 1800 and the standard deviation is equal to 50. So Ashwini is partly right, but we have no further information about the distribution in the population. It is some distribution with those parameters not necessarily normal, but by applying the central limit theorem, we are only able to approximate the distribution of x bar to be normal. Yes, yes, I should remember that. Yes, and as I said before, it is important to keep in mind what the central limit theorem allows us to say and what it does not allow us to say.

All right, let's visit Dr. Sasdev in Jodhpur next. Dr. Naveen Sasdev runs a medical practice in Jodhpur. He has been reading about the duration of Alzheimer's disease in patients. The

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duration from the onset of symptoms until death. I have a close friend whose father was recently diagnosed with Alzheimer's disease. Oh, very sorry to hear that, Tejas. I hope he's getting some good care. Dr. Sazdev has learned that the time duration from the onset to death ranges from 3 years to 20 years of age, with an average of 8 years and a standard deviation of 4 years.

Dr. Sasdev asks his staff to randomly select medical records of 30 diseased Alzheimer patients from their medical records repository and find the average duration corresponding to this sample. He wants his staff to find the probability that the average duration is less than 7 years. He wants to know what is the probability that the average duration exceeds seven years.

And he wants to know what is the probability that the average duration lies within one year of the population mean of it. Now what do we know about the distribution in the population? What can we say about the shape of the sampled population? The duration of the disease range from 3 to 20 years. Yes, and what about the population mean?

It has a population mean of 8 years. Yes, so clearly the population distribution is not symmetric because the mean of 8 does not lie halfway between the maximum of 20 and the minimum of 3. The distribution is most likely skewed to the right with a few patients living a long time after the onset of the disease.

But, regardless of the shape of the population distribution, the sampling distribution of x bar will have a mean of 8 and a standard deviation given by sigma over root n, which would be 4 divided by the square root of 30 in this case. It comes out to be 0. 73. Yes, and in addition, because the sample size is 30, we can use the central limit theorem to say that the distribution of x bar is approximately normal.

Then, to compute the probability that x bar is less than 7, we can use our normal template. We input the value of the mean to be equal to 8 and the standard deviation to be equal to the standard error, which is 0. 73. Next, we look at the cumulative probability section and compute the probability that X bar takes a value of 7 or less.

We get that probability to be equal to 0. 08. There is roughly 8 percent chance that the average duration of the Alzheimer's disease is less than 7 years. Yes, exactly. To compute the

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probability that X bar exceeds 7 years. We want to compute the complement of this event or the tail probability corresponding to 7.

We look at the tail probability section and we compute the relevant probability to be equal to 1 minus 0. 08, which is 0. 92. Next, we want the probability that x bar lies within one year of the population mean of 8. So we want the probability that x bar lies between 7 to 9 years. We have already entered the mean to be 8 and the standard deviation to be equal to 0. 73 in the template. So what do we do next, Ashwini? We enter the value of a to be 7 and b to be 9 and the required probability to be equals to 0. 83.

Well done and I hope these examples give you a good idea about how the sampling distribution of x bar can be applied. Next, We will talk about the sampling distribution of the sample proportion, P bar.

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Sampling Distribution of Proportion

The sample proportion p bar is a point estimator of the population proportion p. The formula for computing the sample proportion p bar is equal to x divided by the sample size n, where x is the number of elements in the sample that possess a characteristic of interest. If we go back to Basavaraju's customer survey, We had computed the point estimate p bar to be equal to 0. 47. Do you remember how we got that and what proportion we were calculating? Basav Raju was interested in the number of customers that had given a satisfaction score of 60 or more. Yes, and we had counted that number in our sample of 36 and we had found x to be equal to 17. So, when we divided 17 by 36, then we got the sample proportion of P bar to be equal to 0. 47. As we saw earlier, the sample proportion P bar is a random variable, and its probability distribution is going to be called the sampling distribution of P bar. To determine how close the sample proportion P bar is to the population proportion P, we need to understand the properties of the sampling distribution of p bar, the expected value of p bar, the standard deviation of p bar, and of course, the shape or the form of the sampling distribution of p bar.

Just like we did for the sampling distribution of x bar. Yes, and we will follow exactly the same steps. Let's first talk about the expected value of p bar. The expected value of p bar corresponds to the mean of all possible values of p bar. We can show that this expected value, this mean, is equal to the proportion p.

In other words, even though the p bar from our sample is a random variable, like x bar, it also comes from a distribution that is at least centered at a good place. p bar comes from a distribution that is centered at p. the value of the proportion p corresponding to the population. So, even though Basaraju got a p bar of 0. 47, which is an instance of a random variable, it comes from a distribution centered at the p corresponding to the population. Because the expected value of p bar is equal to p, we say that p bar is an unbiased estimator of p. Just like how x bar was an unbiased estimator of the population mean mu. Yes, and our estimator p bar also has the same unbiased property.

Just for argument's sake, let us say that Basaraju believes that 50 percent of his customers give a satisfaction score of 60 or more, then we have the proportion in the population p to be equal to 0. 5. And the p bar of 0. 47 that we got from the survey he did recently comes from a

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probability distribution that has a mean or the expected value of 0. 5. Yes, Tejas, you are right. Next, let us talk about the standard deviation of the sampling distribution of p bar. Just as we found the standard deviation of the sampling distribution of x bar, The standard deviation of the sampling distribution of p bar depends on whether the population is finite or infinite.

However, as I did with the sampling distribution of x bar, I will focus mainly on the case where the population size denoted by capital N is large in relation to the sample size small n. And therefore, the result of the expression corresponding to an infinite sample population is large can be applied.

If you recall, we had used the term standard error to refer to the standard deviation of x bar. Do you know why? Because we do not want to confuse it with sigma of the population. Correct. And at that time, I had also said that the general term of standard error is used to refer to the standard deviation of any point estimator.

Yes, I remember that. Therefore, for sample proportions 2, we will use the term standard error of proportion and that will refer to the standard deviation of p bar. We can show that the standard deviation of p bar of the standard error of the population proportions is given by the square root of p times 1 minus p divided by n.

Why is the standard deviation of p bar given by this expression? Yes. I will explain that in a minute when I talk about the shape of the distribution of p bar. Since the standard error of p bar is given by the square root of p times 1 minus p divided by n, Basuraju's p bar of 0. 47 comes from a distribution centered at p is equal to 0. 5 and with a standard deviation or standard error corresponding to the square root of 0. 25 divided by 36, which comes out to be equal to 0. 833. Now, let's talk about the form of the sampling distribution of p bar. What was the shape of the distribution of x bar? It is normal if we are sampling from the population that is normal regardless of the sample size n and can be approximated by the normal due to CLT if the sample size is 30 or more respective of the distribution in the population.

Well done, I am impressed. We are good students, you know. Yes, you are. Yes, you definitely are, and I'm glad to have you. Now, back to the distribution of p bar. We know that the mean

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and the standard deviation of the sampling distribution of p bar are given by p and the square root of p times 1 minus p divided by n, respectively.

To determine the shape and the form of the sampling distribution, we first note that the sample proportion p bar is equal to x over n. Now, for a simple random sample from a large population, the value of X is a binomial random variable. Now, my good students, can you tell me why I say that? In Basavraju's case, X is a random variable corresponding to a number of customers who gave a score of 60 or more.

Yes, and what is the connection then to the binomial random variable? If we define success to be getting a score of 60 or more, then X is counting the number of successes in n samples or n trials. Wonderful. Yes. And as you rightly said, Tejas, we are looking at the number of elements in the sample that satisfy a particular characteristic of interest.

In this case, the sample characteristic of interest is whether a customer gave a score of 60 or more. So now, if x is binomial, what is its mean and standard deviation? Its mean is p, the proportion in the population, and its variance is n times p times 1 minus p. Yes, and if n is large enough, so that n times p and n times 1 minus p are both greater or equal to 5, what do we have?

The normal approximation to the binomial. Yes, absolutely. You are both on a roll today. So now, we have x, the number of successes, or customers who have a score of 60 or more. In Basaraju's case, To be approximated by a normal with a mean of N times P and a variance equal to N times P times one minus P.

Now, if I take this normal random variable and divide it by a constant N, it still remains normal, right? Yes, but it's mean will be N times P divided by N, which is P, and its variance will be N times P times one minus p. divided by n squared. So that gives the standard deviation to be equal to the square root of p times 1 minus p divided by n.

And that was exactly our expression for the standard error of p bar. Oh, that is how we got it. Yes. And so the probability distribution of the sample proportion p bar, the random variable

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that is equal to x over n, can be approximated by a normal distribution. And now we have figured out the shape of the distribution of p bar in addition to its mean and standard error.

This gives us a key insight. In most practical applications, Where sample proportion is to be computed, the sample sizes are almost large enough to permit the use of this normal distribution. This would imply that our sampling distribution of p bar is normal, centered at the population proportion p, and with the standard error sigma p bar equal to the square root of p times 1 minus p divided by n.

The practical value of the sampling distribution of p bar is that it can be used to provide probability information about the difference between the sample proportion and the population proportion. Let's say Basaraju wanted to know the probability of obtaining a value of p bar from his sample that is within 0. 05 of the population proportion. In other words, If we assume that the proportion in the population P is equal to 0. 5, then Basaraju wants to know what is the probability that his sample survey of 36 respondents would give a sample proportion P bar that lies between 0. 45 and 0. 55. Knowing that the sampling distribution of p bar can be approximated by a normal distribution that is centered at the mean of 0. 5 with a standard deviation of sigma p bar of 0. 833, we can find the probability that this normal random variable lies between 0. 45 and 0. 55.

Template time again. Yes, we enter the mean to be equal to 0. 5 and the standard deviation to be equal to 0. 0833 in our template. Entering A to be equal to 0. 45 and B to be equal to 0. 55, we find the required probability to be equal to 0. 452. Now, if Basaraju had got a larger response of let's say 100 responses, that is n equal to 100, what happens? Then p bar would still come from a normal centered at 0. 5. Yes, but the standard error of the population proportion would be equal to the square root of 0. 25 divided by 100, which would be equal to 0. 05. Then, with a sample size of 100, the probability that the sample proportion p bar would lie between 0. 45 and 0. 55 will be equal to 0. 683 from our template. So we see that if the sample size was increased from 36 to 100, the probability of the sample proportion p bar lying within 0. 05 of the population proportion p will increase from 0. 452 to 0. 683. Good.

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Now, let's go to Bikaner. Sure, let us go. Dr. Pradak Chandak manages a pediatric wing of the Marwar Hospital in Bikaner. He sees a report in the Indian Journal of Pediatrics that says that 42 percent of primary care doctors feel that their patients receive unnecessary medical care.

He is curious to know whether these results apply to doctors in his district. He asked the staff of the Marwar Hospital to take a telephonic survey of 150 doctors, selected at random. He wants to know what is the probability that the sample proportion will be within plus or minus 0. 03 of the population proportion.

And he also wants to know what is the probability that the sample proportion will be greater than 0. 4. Let's answer these questions. Let p bar be the proportion of doctors that feel that their patients receive unnecessary medical care. Then we know that this is a random variable. Since the number of samples chosen for the survey is 150 and is large enough, we can now apply the results of the central limit theorem.

We can say that P bar can be approximated by a normal distribution. This normal distribution would have a mean p equal to 0. 42, the claimed proportion in the population. This distribution has a standard error sigma p bar that is given by the square root of p times 1 minus p divided by n. This yields a sigma p bar value equal to the square root of 0. 42 times 0. 58 divided by 150 and that comes out to be equal to 0. 04. Now, for this random variable that is normal with a mean p of 0. 42 and a standard deviation or standard error equal to 0. 04, we want the probability that it takes a value between 0. 39 and 0. 45. We substitute the mean and the standard deviation in the template and use values of 0. 39 as A and 0. 45 as B and read out the probability to be 0. 547. Yes, that is right. Now next, we want the probability that the sample proportion is greater than 0. 5. We compute the tail probability corresponding to 0. 5. It comes out to be 0. 023. Yes, that is right. Good. Now that we are done with this question, are you hungry?

We are next visiting Manohar who is working with Haldiram.

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Examples of Sampling Distribution of Proportion

Manohar Agarwal works at the original sweet and namkeen shop where the story of Haldiram Retail Sweets began. A descendant of the family of Ganga Bishan Agarwal who started this famous chain, Manohar takes great care in the success of his business. He has learned that 17 percent of the households in Rajasthan spend more than 2, 500 rupees per month on retail snacks and sweets.

He surveys 80 of his loyal customers to obtain more details about their monthly spend on snacks and sweets. He wants to know what is the probability that the sample proportion will be within plus or minus 0. 05 of the population proportion. And how would this probability change if he has surveyed 160 customers instead of the 80?

Let us help him. First, let P bar be the proportion of customers that spend more than 2, 500 rupees per month on retail snacks and sweets. We know that P bar is a random variable in the survey done by Manohar. Since the number of customers chosen is 80, we can apply the results from the central limit theorem.

We can say that the distribution of P bar can be approximated by a normal distribution. This normal distribution has a mean of P equal to 0. 17, the claimed proportion in the population. This distribution has a standard error sigma p bar that is given by the square root of p times 1 minus p divided by n.

So what is the value of sigma p bar? This yields a sigma p bar value of square root 0. 17 into 0. 83 divided by 80 which is equals to 0. 042. Now, for this normal random variable with mean p equal to 0. 17 and the standard deviation or standard error equal to 0. 04, we want the probability that it takes values between 0. 12 and 0. 22. We substitute the mean and the standard deviation in the template and use values of 0. 12 as a and 0. 22 as b and read out the probability to be 0. 789. Next, we want to compute the same probability for a sample size of 160 customers. So, the sampling distribution of p bar remains normal. Yes, but the new distribution of p bar has a standard error of the square root of 0. 17 times 0. 83 divided by 160 now, which comes out to be 0. 028 and we want the probability that it takes values between 0. 12 and 0. 22. We

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substitute the mean and the standard deviation in the template and use the values of 0. 12 as A and 0. 22 as B and read out the probability to be equal to 0. 926.

Where are we going now? How about Ajmer? Sunita needs our help. Sunita Kamvar works for Mayur Pustaka Bhandar in Ajmer. The store has hired her as a sales rep to increase the book adoption and purchase by schools and colleges. Historically, Sunita gets a book adoption on 25 percent of her sales calls. When her manager analyzed her sales calls for a month as a sample of all her sales calls, she got a standard error of the proportion to be equal to 0. 0625. Based on this information, can we determine how large was the sample that was used in the analysis? How many sales calls did Sunita make in that month? And what is the probability that Sunita will obtain book adoptions of 30 percent or more in her sales calls during a one month period? Ashwini, help us get started.

We are told that the percentage of book adoption is 25 percent historically. This therefore would correspond to the mean proportion in the population. Yes, and we assume that the manager's analysis corresponds to a random sample from this population. If the number of samples she used in her analysis is large enough, then the distribution of the sample proportion based on the results of the central limit theorem, we can say that P bar can be approximated by a normal distribution.

This normal distribution will have a mean equal to 0. 25 and a standard error sigma p bar that is given to be equal to the square root of p times 1 minus p divided by n. And this gives us a sigma p bar value equal to the square root of 0. 25 times 0. 75 divided by n. And we are told that this standard error is 0. 0625. Yes, and therefore, the sample size used must be equal to 0. 25 multiplied by 0. 75 divided by the square of 0. 0625 and that comes out to be equal to 48. So now we can verify that when n is equal to 48, n times p and n times 1 minus p are indeed greater than 5 in this case.

Next, we want to compute the probability of book adoptions being more than 30 percent. In other words, we want to compute the probability that p bar will take a value 0. 3. We know that p bar has a normal distribution with mean equal to 0. 25 and a standard error sigma p bar equal to 0. 0625. For this normal distribution, we want the tail probability corresponding to 0. 3. We

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update the template and read the probability to be equals to 0. 212. In other words, there is a 21 percent chance that she hits an adoption rate of 30 percent in a month. Yes, that is correct.

Good, but I am still hungry. Have you both eaten Ghevar? No, I have not. Well, then let's go to Johri Bazaar in Jaipur. Laxmi Misthan Bhandar LMB in Johri Bazaar of Jaipur is famous for its sweets, especially the Ghevar. Customers from all over visit this location to taste its delicious dessert, especially its paneer ghevar and malai ghevar. If you don't know, ghevar is a disc shaped Rajasthani sweet with a honeycomb like structure made from ghee, maida, and sugar syrup. It is usually made during festivals like Teej and Raksha Bandhan. You must try it, and we will have some after we solve this problem.

The Food Lovers magazine reports that 75 percent of the customers of LMB prefer the paneer ghevar as opposed to the plain or the malai ghevar. Malival, who works for Lakshmi Mithan Bhandar, wants to verify this and conducts a random sample of 200 of his customers. He asks them to pick their favorite ghevar flavor.

For Based on their responses, Maliwal wants to know what is the probability that the proportion of people that prefer paneer kebab is greater than 80 percent. He also wants to know what is the probability that the proportion of people that prefer paneer kebab is less than 60 percent. In addition, he wants to know within what limits would he expect the sample proportion to be lie about 95 percent of the time.

Let's answer these questions. Let P bar be the proportion of customers that prefer paneer kebab. We know that P bar is a random variable in the survey of 200 customers done by Maliwal. Since the number of customers chosen is 200, we can apply the results from the center limit theorem. We can say that the distribution of P bar can be approximated by a normal distribution.

This normal distribution has a mean of p equals to 0. 75, the claimed proportion in the population. Yes, and this distribution has a standard error sigma p bar given by the square root of p times one minus p divided by n. This yields a sigma p hat value of square root 0. 75 into 0. 25 divided by 200 which is equals to 0. 03. Yes. Now, for this normal random variable, with mean p equal to 0. 75 and standard deviation or standard error equal to 0. 03, we want the probability that it takes a value greater than 0. 8. We substitute the mean and the standard

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deviation in the template and we find the tail probability corresponding to 0. 8. We get 0. 048. That is only a 4. 8 percent chance that more than 80 percent would prefer Paneer Ghevar.

Next, we want the probability that it takes a value less than 0. 6. For the same mean and the standard deviation, we now obtain the cumulative probability corresponding to 0. 6. Yes, and it comes pretty close to zero. This implies that more than 60 percent prefer the paneer keval. Finally, Maliwal wants to know, within what limits would he expect the sample proportion to lie about 95 percent of the time. For the same mean and standard deviation, we now enter 0. 95 in the inverse section. We find that the proportion that prefer paneer ghevar will lie between 0. 69 and 0. 81, 95 percent of the time. Well done, both of you. Come, let's go and have some ghevar.

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Distribution of Sample Variance

Let's now talk a little bit about the sampling distribution of sample variance and sample standard deviation. In fact, I'm only going to talk about the sample variance. We can use the insights derived there to answer questions related to the sample standard deviation. First, let's revisit how we computed the sample variance S squared.

Let's say that our samples x1, x2, and so on till xn are independently selected from a population. Let x bar be the sample average. Then, how do we compute the sample variance s square? First, we compute the distance of each data point xi from the sample average and we take its square. That is xi minus x bar square.

Then, we add them up, that is, we computed the sum of the square deviations and then we divided it by n minus one. We call this expression as the sample variance. Why do we divide by n minus one and not by n? There are multiple justifications and derivations that you will find for this in statistics books.

I will give you a simple and intuitive one. The argument is that when you compute the sample variance using the above expression, you lose One degree of freedom, or DOF in short. Let's look at a simple example to understand this idea of degree of freedom. If I told you that the average of three numbers is 10, and two of these three numbers are 1 and 2 respectively.

Then, Ashwini, what would be the third number? The third number has to be 30 minus 1 minus 2, which is 27. Yes. So, when we say that the sample average of a set of n numbers is some quantity x bar, not all of these n numbers are free to take on arbitrary values. Only n minus 1 of them are actually free, and the last one has to be such that the sample average comes back to x bar. So we say that we have lost one degree of freedom. Therefore, in the numerator of the expression for sample variance, we add the sum of square deviations and not all of these n squared deviations are free. So, only n minus 1 of them are free. Therefore, when we divide, we divide by n minus 1 and not n. That's the reason.

Now that we have seen the expression for sample variance, we can say the following. Like we had with x bar and p bar, the sample variance is also a random variable. Therefore, it must have

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a probability distribution with a mean, with an expectation, and a shape or form. The question is, what are they? If you recall for x bar, the corresponding answers are depended on whether the population had a normal distribution or not.

If the population distribution was not normal, then central limit theorem allowed us to approximate the distribution of x bar when the sample size was large enough. Similarly, for p bar, if the sample size was large enough, Central limit Theum and normal approximations allowed us to approximate the distribution of tbar with a normal distribution.

However, for sample variance, unfortunately without knowing that the population is normally distributed, we are not able to say much about the distribution of the sample variance, not even approximately. Unfortunately, there is no CLT like result for sample variance. But what we do know is the following.

Let's say that our samples x1, x2, and so on until xn are independently selected from a normally distributed population with some unknown population variance given by sigma squared. Let us say that we have computed the sample variance s squared using the expression that we just saw. Then the quantity n minus 1 times s squared divided by sigma squared is also going to be a random variable.

That's because S square is a random variable. And if you multiply it by a constant n minus 1, and divide it by another constant of sigma square, it still remains a random variable. And it can be shown that this random variable, n minus 1 times S square divided by sigma square, has a chi square distribution of with n minus one degrees of freedom.

Now, I'm not going to go into the proof here, but we already saw the general shape of the chi square distribution in the previous module. And what do we know about the chi square distribution? Do you remember? We know it can take only positive values. Yes. And it is not symmetric, usually. Excellent. Yes. And as the degree of freedom increases, it tends to look symmetric and bell shaped.

Observe that our result above does not directly give us the sampling distribution of s square, like we had for p bar and x bar. Instead, it is for s square multiplied by n minus 1 divided by

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the population variance. Now, this might not seem like a very useful result given that it is the distribution of a quantity involving both the estimator s square and the population parameter sigma square.

We will see how this can be utilized to quantify how far s square is from the population parameter sigma square. Let us see how we can apply this in the context of Basaraju's customer satisfaction survey. If you recall, we had 36 responses. that is n is equal to 36. And our sample average x bar was 57. 31. And our sample standard deviation was equal to 6. 38. And our sample variance equal to 40. 73. We have discussed extensively about the distribution of x bar and the distribution of t bar. Now we will use what we just learned about sample variance.To say something about how far Basaraju's estimate of the standard deviation of 6. 38 and his estimate of the sample variance of 40. 73 are from the population standard deviation sigma and the population's variance sigma square. Of course, as we just learned, we will have to assume that Basaraju's samples are coming from a normal population in order to be able to do so. Let us say we are okay to assume that for now.

Then, we know that the quantity n minus 1 times s square divided by sigma square is a chi square random variable with n minus 1 degrees of freedom. So, in Basaraj's case, this quantity would be a chi square distribution with 35 degrees of freedom because our n is equal to 36. Knowing that n minus 1 multiplied by s squared divided by sigma square is a chi square random variable with 35 degrees of freedom, we can estimate how far Basaraju's estimate of variance of 40. 7 is from the population's variance sigma squared.

Ashwini, let's pull out our chi-square template. Let us look at our template for the chi-square random variable and set the degree of freedom to be equal to 35 and see what we can say about this random variable. Let us put in the degrees of freedom equal to 35 in the template.

Then let us use the inverse probability section. Let us find the value of a chi-square 35 for a cumulative probability of 0. 9. What do we get? We get the values of the chi square random variable to be 46. 06. Yes, so therefore, we can say that a chi square 35 takes a value less than 46. 06, roughly 90 percent of the time.

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Therefore, n minus 1 times s square divided by sigma square is less than 46. 06, 90 percent of the time. Now, if you substitute n is equal to 36 and s square equal to 47. 03 from Basaraju's data, we can say that this implies that the sigma square will be greater than 35 times 40. 73 divided by 46. 06, roughly 90 percent of the time.

This yields sigma square greater than or equals to 30. 92, roughly 90 percent of the time. Yes, and taking square roots, this implies that the population sigma is greater than 5. 56, 90 percent of the time. Next, if we look at the tail probability section and do the same, enter the value of 0. 9 as the tail probability in the inverse section for a chi square 35, what do we get? We get the value to be 24. 8. Yes, and so therefore, we can say that a chi-square 35 takes a value greater than 24. 8, 90 percent of the time. Therefore, n minus 1 times s square divided by sigma square is greater than 24. 8, 90 percent of the time. If you substitute n equal to 36, and s square equal to 47. 03 like we did earlier, we can say that this implies that sigma square will be less than 35 times 40. 73 divided by 24. 8, 90 percent of the time. And this yields a sigma square less than 57. 44, 90 percent of the time.

Taking square roots, this implies the population sigma is less than 7.6 roughly 90% of the time. Yes, and so putting these two results together, we have been able to say that based on Basaraju's one sample of 36 respondents and one estimate of S equal to 6. 38, the population standard deviation sigma is more than 5. 56, roughly 90 percent of the time, and less than 7. 6, 90 percent of the time. So even though we are unable to precisely get the distribution of the sample variance and the sample standard deviation, like we did for X bar and P bar, we are still able to use our sample estimates to compute useful probability information about the population standard deviation and population variance.

Let us now revisit some of the examples that we did for the sample mean and look at some relevant variance related questions.

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Examples of Distribution of Sample Variance

If you recall, Kailash Sharma was studying the unemployment data for Rajasthan. He had found the average number of weeks an individual is unemployed is 17. 5 weeks and that the population standard deviation is 4 weeks. Further, Kailash Sharma was proposing a follow up study with a random sample of 50 unemployed individuals. And we want to know what is the probability that this sample would yield a sample variance in excess of 20. And we want to know what is the probability that his sample of 50 will yield a sample standard deviation that is less than 3. Shall we try and answer these questions.

Let's start with the first one. What is the probability that this sample would yield a sample variance more than 20? We know that the population standard deviation is 4 and the sample size is 50. Yes, and let us say that Kailash's sample of 50 yielded a sample variance of s square. Then, n minus 1 times s square divided by sigma square is a chi-square random variable with degree of freedom equal to n minus 1, which is 49 in this case.

Then, the probability of the sample variance s square being greater than 20 is equivalent to the probability that a chi-square random variable with degrees of freedom equal to 49, taking values greater than 49 times 20 divided by 16. That is, we want the probability that a chi-square 49 is greater than 61. 25. Yes, and so now we can go to the chi-square template. We update the degree of freedom to be equal to 49. We enter 61. 25 in the tail probability section, and we read off the probability to be equal to 0. 113. Therefore, there is a 11. 3 percent chance that Kailash sample will yield a variance that is more than 20.

Next, we want to determine the probability that his sample of 50 will yield a sample standard deviation that is less than three. Now let us say that kage sample of 50 yielded a sample standard deviation of S. Yes. Then we know that the population standard deviation is four. and the sample size is 50. Then n minus one times s squared divided by sigma squared is a chi-square random variable with degree of freedom equal to 49. And the probability of the sample standard deviation S being less than three is equivalent to the probability that a chi-square random variable with degree of freedom 49 taking values less than 49 times 9 divided by 16. That is, we want the probability that a chi-square 49 is less than 27. 5625. Yes, so now we go to the chi-square template. We already have the degree of freedom equal to 49. Now we enter 27. 56

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in the cumulative probability section and we read off the probability to be equal to 0. 006. Therefore, there is a very slim chance that Kailash's sample will yield a standard deviation that is less than 3.

Next, let us revisit Motilal Oswal. As you might remember, Motilal Oswal runs a tax consulting practice based out of Udaipur. He saw a TV report that says that the average fee charged for tax preparation is 1800 rupees and that the standard deviation for the fee charged to an individual was 50 rupees. Motilal had asked his staff to choose a sample of 40 tax returns from his past clients. Now, he wants to know what is the probability that the sample will yield a standard deviation of more than 60. Upon requesting his staff to do this exercise, They claim that there is a 90 percent chance that the standard deviation in the sample will be less than 55. Is this claim true? Can we help Motilal with these questions?

Ashwini, your turn now to help us get started. We know that the population standard deviation is 50 and the sample size is 40. Yes, and so let us say that Motilal's sample of 40 yielded a sample standard deviation of s. Then n minus 1 times s square divided by sigma square is a chi-square random variable with degree of freedom equal n minus 1, which is 39 in this case.

The probability then of the sample standard deviation s, being greater than 60 is equivalent to the probability that a chi-square random variable with degree of freedom 39 taking values greater than 39 times 60 square divided by 50 square. That is, we want the probability that a chi-square 39 is greater than 56. 16. We go to the chi-square template. We update the degree of freedom to be equal to 39. We enter 56. 16 in the tail probability section, and we read off the probability to be equal to 0. 037. Therefore, there is a 3. 7 percent chance that Motilal's sample will yield a standard deviation in excess of 50.

Next, we want to test the claim that there is a 90 percent chance that the standard deviation in the sample will be less than 55. Again, let's assume that Motilal's sample of 40 yielded a sample standard deviation of S. Yes, and now we know that the population standard deviation is 50 and the sample size is 40.

Then n minus 1 times S squared divided by sigma squared is a chi-square random variable with 39 degrees of freedom. And then the probability of a sample standard deviation S being less

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than 55 will be equivalent to the probability that a chi-square random variable with degree of freedom 39 taking values less than 39 times 55 square divided by 50 square. That is, we want the probability that a chi-square 39 is less than 47. 19. Yes, we go to the chi-square template. We already have the degree of freedom set to be equal to 49. We enter 47. 19 in the cumulative probability section and we read off the probability to be equal. 0. 827. So there is an 83 percent chance of this happening. It is high, but not as high as the 90 percent that the staff claims.

Next, let us revisit Dr. Naveen Sachdev in Jodhpur. If you recall, Dr. Sachdev had learned that the time duration from the onset to death in patients diagnosed with Alzheimer's ranges from 3 to 20 years of age, with an average of 8 years and a standard deviation of 4 years. Dr. Sachdev had asked his staff to randomly select medical records of 30 diseased Alzheimer patients from their medical repository. And now he wants to know what is the probability that their sample study will reveal a sample standard deviation of less than three years? And what is the probability that their sample study will reveal a sample standard deviation that lies between three and a half and four and a half years?

Let's get cracking on these questions. We know that the population standard deviation is four years and the sample size is 30. Let us say that the study done by doctors as their staff using a sample of 30 resulted in a sample standard deviation of s. Then, n minus 1 times s square divided by sigma square is a chi-square random variable with degrees of freedom equal to n minus 1, which is 29 in this case.

Then, the probability that the sample standard deviation s being less than 3 is equivalent to the probability that a chi-square random variable with degree of freedom 29, taking values less than 29 times, 9 divided by 16. That is, we want the probability that a chi-square 29 is less than 16. 3125. Yes, so now we go to the chi-square template.

We go to the chi-square template. We update the degree of freedom to be equal to 29. We enter 16. 31 in the cumulative probability section. And we read off the probability to be equal to 0.028. Therefore, there is a 2. 8 percent chance that doctors as their staff will find that their sample standard deviation to be less than 3 years.

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Next, we want to determine the probability that the sample standard deviation S that the staff get from their sample study will lie between 3.5 and 4.5 years. Now we know that the population standard deviation is 4 and the sample size is 30. Yes, so therefore, n minus 1 times s square divided by sigma square is a chi-square random variable with degree of freedom equal to 29.

And the probability that the standard deviation s will lie between 3 and a half and 4 and a half is equivalent to the probability that a chi-square random variable with degree of freedom 29 taking values between three and a half square multiplied by 29 divided by 16 and four and a half square multiplied by 29 divided by 16. That is, it takes values between 22. 20 and 36. 70. Yes, and we enter A to be equal to 22.2 and B to be equal to 36. 7 and we read off the relevant probability to be equal to 0. 658. Therefore, there is a 66 percent chance that the sample standard deviation obtained by Dr. Sachdev staff will lie between three and a half and four and a half.

Well done, both of you. I hope that gave you some good feel for the distributions of the sample variance and how we are able to answer some useful questions about the population standard deviation and variance. Remember that all of these results are under the assumption that the population distribution is normal. It could be possible in the case of Kailash Sharma's analysis of unemployment and Motilal's analysis of tax filing charges. That reminds me, we do know that the population distribution in the case of Dr. Sachdev's population is definitely not normal. Yes, that is true. In fact, we know that it is skewed.

So, one should take the conclusions regarding the sample standard deviations with some caution. Is it wrong? Where we wrong in our analysis for Dr. Sachdev? Perhaps as a statistician, one could argue that we were wrong. But as someone wanting to make some decisions, we could argue that even though the model we chose made some inappropriate assumptions it still might be giving Dr. Sachdev some insights that might be useful. What could he do? We might suggest that Dr. Sachdev use a larger sample than 30 to get better results. It is possible that with larger samples, the random variable corresponding to n minus 1 times s squared divided by sigma square at least resembles a chi-square with n minus 1 degrees of freedom, even if it is not exactly a chi-square.

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Properties of Point Estimator

In this module, we have seen how sample statistics such as the sample mean x bar, sample standard deviation s, and the sample proportion p bar can be used as point estimators of their corresponding population parameters mu, sigma, and p respectively. Interestingly, in each of these cases, the sample statistics is the point estimator of the corresponding population parameter.

However, did you wonder why we did not look at point estimators of other statistics, like the median, the quartile, the skewness, etc.? Yes. Why did we focus on these? A couple of reasons. First, of course, the ones we looked at are very common and useful. So we must look at them. Second, these point estimators have good properties also.

What kind of properties are we looking for in these point estimators? Before using a particular sample statistics as a point estimator, usually we would want to check whether the simple statistics demonstrate certain properties associated with a good point estimator. Therefore, what I'm going to talk next is about some of the general properties that we would expect of a good point estimators. In particular, we will focus on four desirable properties, namely point estimators being unbiased, efficient, consistent, and sufficient. Because our discussion is going to be in general terms, and not about a particular point estimator like x bar or p bar, and different sample statistics can be used as point estimators for different population parameters.

We will follow a slightly different notation. A new Greek letter enters. We will denote by the Greek letter of theta, the population parameter of interest. For example, theta could very well correspond to either mu, or sigma, or p, or any other population parameter of interest. Similarly, we will use the Greek letter theta bar to denote the sample statistic or the point estimator of the corresponding population parameter theta.

Again, this could be x bar, p bar, or s, or any other sample statistics. Now, let's look at each of those four properties in detail. Let us start by looking at one of the main properties of a point estimator, it being unbiased. We have already referred to this property when we talked about x bar and p bar.

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In this discussion, we are going to generalize this concept. If the expected value of a sample statistic is equal to the population parameter being estimated, then the sample statistic is said to be an unbiased estimator of the population parameter. In other words, if a sample statistic is an unbiased estimator of the population parameter theta, if the expected value of theta bar is equal to theta.

Here the expected value of mean of all possible values like expected values of x bar and t bar. Yes, exactly. The expected value of an unbiased sample statistic is equal to the population parameter being estimated. In this figure, we can see cases of unbiased and biased point estimators. The figure shows that for an unbiased estimator, the mean of the sampling distribution is equal to the value of the population parameter.

In other words, the estimation errors balance out in this case, because sometimes the value of the point estimator theta bar might be less than theta, and at other times it may be greater than theta. However, in the case of a biased estimator, the mean of the sampling distribution is either strictly less than or strictly greater than the value of the population parameter.

In this picture, you can see the expected value of theta bar is different from the value of theta. That is, the sample statistic has a high probability of mis-estimating the value of the population parameter. The amount of bias is also shown in this figure. So what happens if the estimator is biased? Do we just not use such an estimator?

Not really. Having a biased estimator is not necessarily an unsound mountable problem. It is a problem, but if the bias in the estimator can actually be quantified, then a bias correction can be done. But people usually prefer to work with an unbiased estimator instead. For example, if you recall that when we discuss sampling distributions of both the sample mean and the sample proportion.

We had stated that the expected value of X bar is equal to the population mean mu, and the expected value of P bar is equal to the population proportion P. In both cases, X bar and P bar are unbiased estimators of their corresponding population parameters mu and P. Also, in the case of the sample standard deviation, S, and the sample variance, S squared, it can be shown that the expected value of E of S squared is equal to the population variance sigma squared.

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In fact, when we looked at the formulas for sample variance, we had used a value of n minus 1 rather than n in the denominator. You mean in the formula where we were adding the sum of squared deviation and dividing by n minus 1. Yes, in that formula. And at that time I had given the reason for using n minus 1 rather than n.

And we said that it was to make the sample variance an unbiased estimator of the population variance. Oh, so if we can, we try and correct for the bias in the estimator? Yes, exactly. Now, let's talk about the second property desired from point estimators, efficiency. If a simple random sample can be used to provide two unbiased point estimators for the same population parameter theta, say mu or p, then in this situation, the point estimator with a smaller standard error is preferred, right?

Yes, because it tends to provide estimates that are likely to be closer to the population parameter. Exactly. And the point estimator with a smaller standard error is said to have a greater relative efficiency than the other. In the figure, we see sampling distributions of two unbiased point estimators, theta bar 1 and theta bar 2.

They are unbiased, as they are both centered at theta. The standard error of theta bar 1 is less than the standard error of theta bar 2. Therefore, the values of theta bar 1 have a greater chance of being closer to the parameter theta that we are trying to estimate. It appears that the standard error of point estimator theta bar 1 is less than the standard error of point estimator theta bar 2.

Yes, it is. And theta bar 1 is said to be relatively more efficient than theta bar 2 and that would be our preferred point estimator. Now, Tejas and Ashwini, if you recall, we had talked about how the median is robust to outliers in the data. However, one disadvantage of the sample median is that even if you are sampling from a normal population, the standard error of the sample mean is much less than the standard error of a sample median. In other words, the sample mean is a more efficient estimator than the sample median.

The third property desired from point estimators is consistency. A point estimator is said to be consistent if the values of the point estimator tend to be closer to the population parameter as the sample size becomes larger. In other words, a large sample size tends to provide a better point estimate than a small sample size. For example, when we looked at the sample mean x

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bar, we showed that the standard error of x bar is given by sigma over root n. Now, because sigma x bar is related to the inverse of the square root of n, the sample size, larger sample sizes provide smaller values of sigma x bar.

Therefore, we can conclude that a larger sample size tends to provide point estimates that are closer to the population mean mu. So, we can say that the sample mean x bar is a consistent estimator of the population mean mu. Yes, and similarly, when we looked at the sample proportion p bar, we showed that the standard error of p bar, which is sigma p bar, is given by the square root of p times 1 minus p divided by n.

Again, because sigma p bar is related to the sample size with n in the denominator, the larger the sample size, the smaller are the values of sigma p bar. Therefore, a larger sample size will provide better point estimates, estimates that are closer to the population parameter p.

Next, we discuss another desirable property of point estimators. A point estimator is said to be sufficient if it contains all the information in the data about the parameter estimates. That is, we say that an estimator uses all the available data in the sample of size n in order to compute the estimator. If you recall, when we computed the sample mean x bar, we took all the data points in the sample and divided by the sample size n.

Similarly, when we looked at the sample proportion p bar, we took all the samples data points and identified the ones that satisfied our desired criteria, the case of 60 or more if you remember. and divided that x by n to get our sample proportion. Therefore, the sample proportion p bar and the sample mean x bar are sufficient estimators.

Because when we computed both of these estimators, we consider the entire data set. Yes, exactly. But a sample median would not be considered as a sufficient estimator. So, in conclusion, a couple of points about why we, as much as other practitioners of statistical inference, are interested in the specific sample point estimators like X bar, P bar, and S.No doubt, mean, variance, and proportion are fundamental properties of a population and a sample. Understanding what they are gives us a good sense of the sample data and the population. But in addition, these point estimators also have very attractive properties in terms of being unbiased, efficient, sufficient, and consistent.

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Sampling Error vs Non sampling errors

Before we close this module, I would like to compare sampling errors and non sampling errors. So far, we have been discussing the purpose of statistical inference is to use sample data to quickly and inexpensively gain insight into some characteristics of the population for which we want to make some decisions. We recognize that individual samples are individual samples. They are all, to some degree, not perfectly a representation of the population. There are going to be some errors. In general, there are two types of errors. Sampling errors and non sampling errors. Sampling error refers to the deviations of the sample from the population that occur due to the random nature of the sample collected.

If repeated random samples of the same size are collected from the population of interest using a probabilistic sampling technique, then on an average, we expect that the sample should be representative of the population, right? Yes, we do. And in fact, this is indeed the justification for collecting sample data in a random manner.

Yes, but there is an underlying luxury of repetition that is assumed in that guarantee. However, random collection of sample data alone does not ensure that a single sample will be perfectly representative of the population of interest, primarily because we have the resources to collect only one single random sample.

Therefore, sampling error is unavoidable and this is the risk that we must accept when we choose to collect a simple random sample. That is why we wanted our estimators to have good properties, and we quantified the sampling errors using standard errors, x bar, p bar, and so on. We can be fairly assured that if we take an extremely large sample, there may be very little potential for sampling errors.

Yes, and similar to sampling errors, there are non sampling errors. To take steps to reduce non sampling errors, we must first understand what non sampling errors are. A sample may fail to represent a target population because the sample collected is very different from the target population for reasons other than random sampling.

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Such errors are known as non sampling errors. These are of three types, coverage error, non response error, and measurement error. Let's see each one of them. The first type of non sampling error that could occur is known as coverage error. To understand what coverage error is, let's consider Basaraju's survey of his current customers at Chroma. Basaraju's objective of measuring satisfaction was to find ways to grow sales revenue. Then, there may be some non sampling errors that Basaraju is making. For instance, his survey that he has conducted will only help him grow sales with his current customers. However, there may be larger sales growth opportunity with customers of Chroma's competition, with customers of his competition that may be not entirely satisfied with the service that they are getting.

So, should Basavaraju include them in his sample too? In fact, one could also argue that there might be an untapped market, people that are neither customers of Basavaraju or his competitors, should they be in the population? Should they be in the sample? The main point here is that if the research objective and the population from which the sample is being drawn are not aligned.

Then, we have what is known as a coverage error. And if you make such errors, the data that Basavaraju collects, no matter how well done through a random sampling, will not help chroma achieve its real purpose. This type of error is often referred to as coverage error.

Next, even when the sample is taken from an appropriate population. Non sampling errors can occur when some of the segments of the target population are systematically under represented or over represented in the sample. This is known as a non response error. For example, suppose Basavaraju collected the data from his existing customers by distributing a form when they came to the store.

Then, although he is trying to collect the survey from his current customers, he might be completely ignoring the customers that might be ordering online. And that could lead to under representation of a certain important segment of his customers. Yes, of course. And alternatively, if he did a purely online survey, then he could be guilty of under representing customers that are coming to the store and making purchases.

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Further, even if he collects data through both online and in store mechanisms, the corresponding response rates might not be reflective of the proportion of customers that order online and in store. These kind of errors, where the study design is flawed because some segments of the population are either more or less likely to respond, is known as non response error.

The third type of non sampling error that could occur is known as measurement error. These types of errors occur if the process of collecting the data, especially when it is done through a questionnaire, are either ambiguous or difficult to respond or understand. Sometimes the responses may not accurately reflect how the respondents intended to respond.

Yes, for instance, in Basavaraju's case, if the form filled by the customer in the store is not anonymous, then a customer might not be honest in their responses. Yes, and alternatively, if the store staff were looking over the customer while he or she were filling the responses, it is likely that the responses are not truly reflective of their sentiments.

Similarly, if the survey is done online, responses might be influenced by the way the online website or app presents these questions. If the form is too long or has confusing questions, then the responses might be ambiguous or incomplete. These types of errors might lead to responses that may or may not be totally reflective of the responses that the respondent intends to give.

Sometimes, people may provide incorrect answers deliberately because the respondents have a poor recollection of facts, or they are unwilling to respond honestly, or they just want to complete the survey in a hurry. These types of errors are known as measurement errors. So, how do we reduce these non sampling errors?

Non sampling errors can introduce bias in the estimates produced from a sample, and this can mislead decision makers in their decision making process. Therefore, no matter how large or how small a sample is, and how much care we have taken to ensure its randomness, care must be taken to minimize the introduction of non sampling errors as well.

Here are some steps that can be taken to reduce non sampling errors. First, we must carefully define the target population before we collect sample data, so that the probability sample is

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drawn from this carefully designed target population. Second, we must carefully design the data collection process and train the data collectors about potential errors.

Third, we must pre test the data collection procedure using a pilot, for instance. To identify and correct for potential sources of non sampling errors prior to the final data collection. Fourth, we must choose the appropriate method among stratified random sampling, cluster sampling, or systematic sampling as the case may be, ensuring that the variables of interest are adequately reflected in the sample.

Hopefully, with some of these measures, errors can be minimized, both sampling and non sampling, so that the inferences are reliable and useful. Now, let us summarize what we have accomplished in this module and set the stage for what we are going to do in subsequent modules. We made an important leap in this module.

In earlier modules, we had talked about sample data by themselves and probability distributions by themselves. In this module, we brought both of them together. We introduced the notion of sample data and the associated probability distributions, which we called as sampling distributions. These form the foundations of statistical inference.

And that's what we're going to be focusing on in the subsequent parts of this course. We saw that sample statistics are different from population parameters. When you collect data and you compute descriptive statistics, in most cases, these are known as sample statistics. And the purpose of computing these sample statistics is to is usually twofold.

First, we want to understand the quality and the nature of the data you have received. But second, and perhaps more importantly, the purpose is to identify what the sample data is telling you about the parameters in the population. The second purpose is more important because the analysis of data is usually motivated by the need to make a decision that affects not just the sample, but the population at large.

And hopefully you saw that with several examples that we discussed how we are able to achieve this. We also looked at several sampling methods. How do you collect the data to ensure that

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the sample that you collect, even if small in size compared to the population, is fairly representative of what the population contains.

We saw how the sample statistics and point estimators can be used to infer useful properties about population parameters. To achieve this, we use the help of a powerful and important result in statistics known as the central limit theorem. This central limit theorem helped us to leap from the sample statistics to the population parameters.

Although we cannot precisely say what these population parameters are going to be from a small sample that is drawn randomly from a population. We can nevertheless provide very specific probability related statements about where the population parameters are likely to be based on just a single sample that we have extracted from the population.

We saw specific applications in the case of x bar, p bar, and s. Finally, we looked at the properties that make some of these point estimators more attractive than the others. In the next module, we will continue to build on this idea of moving on from the sample to the population. We will use other techniques of statistical inference to derive confidence intervals and test hypothesis related to the population parameters.

The intent throughout is to see how through simple sample data and some powerful statistical tools, impactful decisions can be made relevant to the population at large. Thank you Ashwini and Tejas and thanks to you too for your cooperation. I will see you next time.

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